Flows on the Circle: Oscillation and Bottlenecks

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Definition: Circle as a quotient

The **circle** is

$$S^1 = \mathbb{R}/(2\pi\mathbb{Z}),$$

so θ and $\theta + 2\pi k$ represent the same point for every $\theta \in \mathbb{R}$ and $k \in \mathbb{Z}$. The equivalence class of θ is denoted by $[\theta] \in S^1$.

Definition: Vector field on the circle

A **continuously differentiable vector field on** S^1 is represented by

$$\dot{\theta} = f(\theta), \qquad f(\theta + 2\pi) = f(\theta).$$

Here $f : \mathbb{R} \to \mathbb{R}$ is continuously differentiable and 2π-periodic.

Definition: Lift of a circle-valued solution

Let $q : \mathbb{R} \to S^1$ be the quotient map $q(\theta) = [\theta]$, and let $\gamma : I \to S^1$ be a continuous solution curve. A **lift of** γ is a continuous function $\tilde{\theta} : I \to \mathbb{R}$ satisfying

$$q(\tilde{\theta}(t)) = \gamma(t) \qquad \text{for every } t \in I.$$

For the circle vector field represented by $\dot{\theta} = f(\theta)$, the lift is differentiable and satisfies

$$\dot{\tilde{\theta}} = f(\tilde{\theta})$$

after its initial representative has been chosen.

Definition: Fixed point of a circle flow

A class $[\theta^*] \in S^1$ is a **fixed point** of $\dot{\theta} = f(\theta)$ if

$$f(\theta^*) = 0.$$

Question: Which formulas define circle flows?

Analyze

$$\dot{\theta} = \sin\theta \qquad \text{and} \qquad \dot{\theta} = \theta.$$

Solution. The first right-hand side is 2π-periodic. In one period its fixed points are 0 and π, and

θ	$(0, \pi)$	π	$(\pi, 2\pi)$	$0 \equiv 2\pi$
$\sin\theta$	$+$	0	$-$	0

Thus $[\pi]$ is asymptotically stable and $[0]$ is unstable.
The formula $f(\theta) = \theta$ is not 2π-periodic. The equivalent labels 0 and 2π prescribe different velocities, so it does not define a vector field on S^1.

Theorem: Fixed-point–rotation dichotomy

Let $f : \mathbb{R} \to \mathbb{R}$ be continuously differentiable and 2π-periodic.
If f has a zero, every nonfixed trajectory remains in one component of

$$S^1 \setminus \{[\theta] : f(\theta) = 0\}$$

and approaches a boundary fixed point in forward time.
If f has no zero, then every trajectory rotates forever in the same direction and is periodic on S^1.

Proof. If f has no zero, continuity and connectedness imply that f has one sign. Compactness gives $|f(\theta)| \geq m > 0$, so every lift completes a turn in finite time.
If zeros occur, uniqueness prevents a nonfixed trajectory from meeting a fixed point in finite time. On each component of the complement of the zero set, f has one sign. The lift is monotone and bounded by the component's endpoints, so it has a limit L. If $f(L) \neq 0$, continuity would keep the speed bounded away from zero near L, contradicting convergence. Hence $f(L) = 0$.

Proposition: Single-valued potential criterion

Let f be continuous and 2π-periodic. There exists a differentiable, 2π-periodic function V such that

$$f(\theta) = -V'(\theta)$$

if and only if

$$\int_0^{2\pi} f(\theta)\, \mathrm{d}\theta = 0.$$

Proof. If $f = -V'$ and V is periodic, then

$$\int_0^{2\pi} f(\theta)\, \mathrm{d}\theta = -V(2\pi) + V(0) = 0.$$

Proposition: Single-valued potential criterion (continued)

Conversely, if the integral is zero, define

$$V(\theta) = -\int_0^\theta f(s)\,\mathrm{d}s.$$

Then $V' = -f$ and

$$V(\theta + 2\pi) - V(\theta) = -\int_\theta^{\theta+2\pi} f(s)\,\mathrm{d}s = 0.$$

Thus V is a function on S^1.

Definition: Uniform oscillator

A **uniform oscillator** is a circle flow of the form

$$\dot\theta = \omega,$$

where $\omega \in \mathbb{R}$ is constant. Its lifts are

$$\theta(t) = \theta_0 + \omega t.$$

Notation: Angular speed

For the uniform oscillator $\dot\theta = \omega$, its **angular speed** is $|\omega|$. The sign of ω determines the direction of rotation.

Observation: Period of a uniform oscillator

If $\omega \neq 0$, the period of $\dot\theta = \omega$ on S^1 is

$$T = \frac{2\pi}{|\omega|}.$$

Definition: Phase difference

For two phases $[\theta_1], [\theta_2] \in S^1$, their **phase difference** is

$$\phi = [\theta_1 - \theta_2] \in S^1.$$

Question: When does one oscillator lap another?

Suppose

$$\dot\theta_1 = \omega_1, \qquad \dot\theta_2 = \omega_2, \qquad \omega_1 > \omega_2 > 0.$$

Find the lap time. Then evaluate it when the periods are $T_1 = 3$ seconds and $T_2 = 4$ seconds.

Solution. For the lifted phase difference $\tilde\phi = \tilde\theta_1 - \tilde\theta_2$,

$$\dot{\tilde\phi} = \omega_1 - \omega_2.$$

One lap occurs when $\tilde\phi$ increases by 2π, so

$$T_{\mathrm{lap}} = \frac{2\pi}{\omega_1 - \omega_2}.$$

Question: When does one oscillator lap another? (continued)

Since $\omega_j = 2\pi/T_j$,

$$T_{\text{lap}} = \left(\frac{1}{T_1} - \frac{1}{T_2}\right)^{-1} = \left(\frac{1}{3} - \frac{1}{4}\right)^{-1} = 12 \text{ seconds.}$$

Question: How does a nonuniform oscillator change with its parameter?

Let $\omega > 0$, $a \geq 0$, and

$$\dot{\theta} = \omega - a \sin\theta.$$

Classify the dynamics for $a < \omega$, $a = \omega$, and $a > \omega$.

Solution. For $a > \omega$, define

$$\alpha = \arcsin\left(\frac{\omega}{a}\right) \in \left(0, \frac{\pi}{2}\right).$$

parameter	fixed points	dynamics
$0 \leq a < \omega$	none	$\dot{\theta} > 0$; counterclockwise rotation
$a = \omega$	$\theta^* = \pi/2$	half-stable saddle-node point
$a > \omega$	$\alpha,\ \pi - \alpha$	α asymptotically stable, $\pi - \alpha$ unstable

Indeed,

$$f'(\theta) = -a\cos\theta,$$

so $f'(\alpha) < 0$ and $f'(\pi - \alpha) > 0$. At $a = \omega$,

$$f\left(\frac{\pi}{2} + u\right) = \omega(1 - \cos u) \geq 0.$$

More precisely,

$$f\left(\frac{\pi}{2} + u\right) = \frac{\omega}{2}u^2 + O(u^4) \qquad (u \to 0).$$

Locally, the arrows point toward the fixed point from the left and away from it on the right. A point starting immediately to the right nevertheless travels counterclockwise around the remainder of the circle and approaches $[\pi/2]$ from the left. Thus $[\pi/2]$ attracts every initial state on S^1, so it is globally attracting, but it is Lyapunov unstable.

Visualization: Nonuniform-oscillator phase portraits

$0 \leq a < \omega$	$a = \omega$	$a > \omega$
counterclockwise rotation	$\longrightarrow\ \odot_{\pi/2}\ \longrightarrow$	$\longrightarrow\ \bullet_\alpha\ \longleftarrow\ \circ_{\pi-\alpha}\ \longrightarrow$
$\min\dot{\theta} = \omega - a > 0$	$\dot{\theta} \sim \frac{\omega}{2}(\theta - \pi/2)^2 \quad (\theta \to \pi/2)$	$\sin\alpha = \omega/a$

Here $\bullet$ denotes an attracting fixed point and $\circ$ a repelling fixed point. The symbol $\odot$ denotes the threshold point, whose local arrows attract from the left and repel to the right. On the circle, the right-side trajectories travel counterclockwise around the remainder of the circle and approach the point from the left, so this point is globally attracting but Lyapunov unstable.

Theorem: Period of a rotating circle flow

Let $f : \mathbb{R} \to \mathbb{R}$ be continuously differentiable, 2π-periodic, and satisfy $f(\theta) \neq 0$ for every $\theta \in \mathbb{R}$. Every orbit on S^1 has the same least positive period

$$T = \int_0^{2\pi} \frac{\mathrm{d}\theta}{|f(\theta)|}.$$

Proof. If $f > 0$, separation of variables gives

$$T = \int_{\theta_0}^{\theta_0+2\pi} \frac{\mathrm{d}\theta}{f(\theta)} = \int_0^{2\pi} \frac{\mathrm{d}\theta}{f(\theta)},$$

where periodicity makes the value independent of θ_0. If $f < 0$, a lift decreases by 2π in one turn; reversing the limits gives the formula with $|f|$. A return on S^1 requires the monotone lift to change by $2\pi k$ for some nonzero $k \in \mathbb{Z}$, so one turn gives the least positive return time.

Question: What is the exact period of the nonuniform oscillator?

Assume $\omega > a > 0$ and evaluate

$$T = \int_{-\pi}^{\pi} \frac{\mathrm{d}\theta}{\omega - a\sin\theta}.$$

Solution. Use

$$u = \tan\frac{\theta}{2}, \qquad \sin\theta = \frac{2u}{1+u^2}, \qquad \mathrm{d}\theta = \frac{2\,\mathrm{d}u}{1+u^2}.$$

Then

$$\begin{aligned} T &= 2\int_{-\infty}^{\infty} \frac{\mathrm{d}u}{\omega u^2 - 2au + \omega} \\ &= \frac{2}{\omega}\int_{-\infty}^{\infty} \frac{\mathrm{d}u}{(u - a/\omega)^2 + (\omega^2 - a^2)/\omega^2} \\ &= \frac{2\pi}{\sqrt{\omega^2 - a^2}}. \end{aligned}$$

With $\omega > 0$ fixed, as $a \to \omega^-$,

$$T = \frac{2\pi}{\sqrt{(\omega - a)(\omega + a)}} \sim \pi\sqrt{\frac{2}{\omega}}(\omega - a)^{-1/2}.$$

Question: Why does a saddle-node bottleneck produce a square-root delay?

Let $r > 0$ and consider the bottleneck normal form

$$\dot{x} = r + x^2.$$

Define the passage time from $x = -\infty$ to $x = \infty$ as the limit, as $L \to \infty$, of the time required to travel from $x = -L$ to $x = L$. Find this passage time and recover its scaling from a rescaling.

Solution. Separation gives

$$T_{\text{bottle}} = \int_{-\infty}^{\infty} \frac{\mathrm{d}x}{r + x^2}.$$

With $x = \sqrt{r} \tan u$,

$$T_{\text{bottle}} = \frac{1}{\sqrt{r}} \int_{-\pi/2}^{\pi/2} \mathrm{d}u = \frac{\pi}{\sqrt{r}}.$$

Alternatively, set

$$x = r^{1/2} X, \qquad t = r^{-1/2} \tau.$$

Then

$$\frac{\mathrm{d}X}{\mathrm{d}\tau} = 1 + X^2,$$

which contains no r. Its passage time in τ is therefore independent of r. Since $t = r^{-1/2}\tau$, the passage time in t is proportional to $r^{-1/2}$.

Question: How does a flatter bottleneck change the delay?

Let $n \in \mathbb{N}$ satisfy $n \geq 1$, let $r > 0$, and

$$\dot{x} = r + x^{2n}.$$

Determine the dependence of the passage time from $x = -\infty$ to $x = \infty$ on r.

Solution. The rescaling

$$x = r^{1/(2n)} X, \qquad t = r^{1/(2n)-1} \tau$$

gives

$$\frac{\mathrm{d}X}{\mathrm{d}\tau} = 1 + X^{2n}.$$

Consequently,

$$T_{\text{bottle}} = r^{1/(2n)-1} \int_{-\infty}^{\infty} \frac{\mathrm{d}X}{1 + X^{2n}}.$$